DAFFODIL INTERNATIONAL UNIVERSITY

LAB REPORT

DEPARTMENT OF COMPUTING AND INFORMATION SYSTEM

LAB REPORT-01

Course Title : Structured Programming

Course Code : CIS 115

Lab Report : Input Output, Operator

Date : 21-09-2015

Submitted By :

Name: Rashadul Islam

ID: 253-16-019

Section: 23A

Semester: 1st, FALL 2025

Submitted To:

Md. Faruk Hosen

Senior Lecturer,

Department of CIS

Submission Date: 26/09/2025

Submission time: 12:30 pm

Input Output

Problem No: 01

Problem Name: Write a program that prints your name.

Source Code:

```
#include<stdio.h>
int main(){
        printf("My Name is Rashadul Islam");
        return 0;
}
```

Output:

```
PS H:\Programming\c\Basic C> cd "h:\Programming\c\Basic C\exerise class work\1\"
; if ($?) { gcc code1.c -o code1 } ; if ($?) { .\code1 }
My Name is Rashadul Islam
PS H:\Programming\c\Basic C\exerise class work\1>
```

Problem No: 02

Problem Name: An integer variable n contains 5. Write a program that prints the value of "n".

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    n=5;
    printf("the n is :%d",n);
    return 0;
}
```

Output:

```
PS H:\Programming\c\Basic C> cd "h:\Programming\c\Basic C\exerise class work\1\"
; if ($?) { gcc code2.c -o code2 } ; if ($?) { .\code2 }
the n is :5
PS H:\Programming\c\Basic C\exerise class work\1>
```

Problem No: 03

Problem Name: Write a program that read and display an integer number.

Source Code:

```c
#include<stdio.h>
int main(){
    int n;
    printf("Enter a number:");
    scanf("%d",&n);
    printf("the num is :%d",n);
    return 0;
}
```

Output:

```
PS H:\Programming\c\Basic C> cd "h:\Programming\c\Basic C\exerise class work\1\"
; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCo
deRunnerFile }
Enter a number:1099
the num is :1099
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 04

Problem Name: Write a program that read and display an integer number.

Source Code:

```c
#include<stdio.h>
int main(){
    int n;
    printf("Enter a number:");
    scanf("%d",&n);
    printf("the num is :%d",n);
    return 0;
}
```

Output:

```
PS H:\Programming\c\Basic C> cd "h:\Programming\c\Basic C\exerise class work\1\"
; if ($?) { gcc code3.c -o code3 } ; if ($?) { .\code3 }
Enter a number:13
the num is :13
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 05

Problem Name: Write a program that read and display floating point number.

Source Code:

```c
#include<stdio.h>
int main(){
    float n;
    printf("Enter a float number:");
    scanf("%f",&n);
    printf("the float num is:%.2f",n);
    return 0;
}
```

Output:

```
PS H:\Programming\c\Basic C> cd "h:\Programming\c\Basic C\exerise class work\1\"
; if ($?) { gcc code5.c -o code5 } ; if ($?) { .\code5 }
Enter a float number:5.6778
the float num is :5.68
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 06

Problem Name: Write a program that read and display long Number.

Source Code:

```
#include<stdio.h>
int main(){
    long int n;
    printf("Enter a number:");
    scanf("%ld",&n);
    printf("the num is :%ld",n);
    return 0;
}
```

Output:

```
PS H:\Programming\c\Basic C> cd "h:\Programming\c\Basic C\exerise class work\1\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a number:1012456
the num is :1012456
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 07

Problem Name: Write a program that read and display double number.

Source Code:

```
#include<stdio.h>
int main(){
    double n;
    scanf("%lf",&n);
    printf("the num is :%lf",n);
}
```

Output:

```
> cd "h:\Programming\c\Basic C\exerise class work\1\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
12.23565
the num is :12.235650
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 08

Problem Name: Write a program that read and display any character.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("Enter a Character:");
    scanf("%C",&n);
    printf("the character is :%c",n);
}
```

Output:

```
                                                 > cd "h:\Program
ming\c\Basic C\exerise class work\1\" ; if ($?) { gcc tempCodeRu
nnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerF
ile }
Enter a Character:R
the character is :R
PS H:\Programming\c\Basic C\exerise class work\1> 
```

Daffodil International University

Problem No: 09

Problem Name: Write a program that read ASCII value and display equivalent character.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("Enter a ASCII number :");
    scanf("%d",&n);
    printf("the character is :%c ",n);
    return 0;
}
```

Output:

```
> cd "h:\Programming\c\Basic C\exerise class work\1\" ; if ($?) { gcc code9.c -o
 code9 } ; if ($?) { .\code9 }
Enter a ASCII number :69
the character is :E
PS H:\Programming\c\Basic C\exerise class work\1>
```

Problem No: 10

Problem Name: Write a program that read any character and display ASCII value.

Source Code:

```
#include<stdio.h>
int main(){
    char n;
    printf("Enter a character :");
    scanf("%c",&n);
    printf("the ASCII value is:%d",n);
    return 0;
}
```

Output:

```
> cd h:\Program
ming\c\Basic C\exerise class work\1\" ; if ($?) { gcc code10.c -
o code10 } ; if ($?) { .\code10 }
Enter a character :R
the ASCII value is :82
PS H:\Programming\c\Basic C\exerise class work\1>
```

Problem No: 11

Problem Name: Write a program that read any lower-case character and display in upper case.

Source Code:

```c
#include<stdio.h>
int main(){
    char n;
    printf("Enter a LOWER case character:");
    scanf("%c",&n);
    printf("the Upper case charater is:%c",
           n-32);
    return 0;
}
```

Output:

```
> cd  H:\Programmi
ng\c\Basic C\exerise class work\1\" ; if ($?) { gcc code11.c -o co
de11 } ; if ($?) { .\code11 }
Enter a LOWER case character:g
the Upper case charater is:G
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 12

Problem Name: Write a program that read any upper-case character and display in lower case.

Source Code:

```
#include<stdio.h>
int main(){
    char n;
    printf("Enter a upper case character:");
    scanf("%c",&n);
    printf("the lower case charater is:%c",n+32);
    return 0;
}
```

Output:

```
                                         > cd  H:\Programmi
ng\c\Basic C\exerise class work\1\" ; if ($?) { gcc tempCodeRunner
File.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a upper case character:R
the lower case charater is:r
PS H:\Programming\c\Basic C\exerise class work
```

Go to Line/Column

Daffodil International University

Problem No: 13

Problem Name: Write a program that read any decimal number and display equivalent octal number.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("enter a decimal number:");
    scanf("%d",&n);
    printf("the octal num is:%o",n);
    return 0;
}
```

Output:

```
                                              > cd "h:\Programmi
ng\c\Basic C\exerise class work\1\" ; if ($?) { gcc tempCodeRunner
File.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
enter a decimal number:10
the octal num is:12
PS H:\Programming\c\Basic C\exerise class work\1>
```

Problem No: 14

Problem Name: Write a program that read any decimal number and display equivalent hexadecimal number.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("Enter a decimal num:");
    scanf("%d",&n);
    printf("the hexa-decimal is:%X",n);
    return 0;
}
```

Output:

```
                                                  > cd "h:\Programmi
ng\c\Basic C\exerise class work\1\" ; if ($?) { gcc code14.c -o co
de14 } ; if ($?) { .\code14 }
Enter a decimal num:44
the hexa-decimal is:2C
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 15

Problem Name: Write a program that read any decimal number and display equivalent hexadecimal number.

Source Code:

```c
#include<stdio.h>
int main(){
    int n;
    printf("Enter a decimal num:");
    scanf("%d",&n);
    printf("the hexa-decimal is:%X",n);
    return 0;
}
```

Output:

```
                                                  > cd "h:\Programmi
ng\c\Basic C\exerise class work\1\" ; if ($?) { gcc code14.c -o co
de14 } ; if ($?) { .\code14 }
Enter a decimal num:10
the hexa-decimal is:A
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 16

Problem Name: Write a program that read any octal number and display equivalent decimal number.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("Enter a octal num:");
    scanf("%o",&n);
    printf("the decimal is:%d",n);
    return 0;
}
```

Output:

```
                                                      > cd  h:\Programmi
ng\c\Basic C\exerise class work\1\" ; if ($?) { gcc tempCodeRunner
File.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a octal num:44
the decimal is:36
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 17

Problem Name: Write a program that read any hexadecimal number and display equivalent decimal number.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("Enter a hexa-decimal num:");
    scanf("%x",&n);
    printf("the decimal is:%d",n);
    return 0;
}
```

Output:

```
> cd "h:\Programmi
ng\c\Basic C\exerise class work\1\" ; if ($?) { gcc code17.c -o co
de17 } ; if ($?) { .\code17 }
Enter a hexa-decimal num:F
the decimal is:15
PS H:\Programming\c\Basic C\exerise class work\1>
```

Problem No: 18

Problem Name: Write a program that read and display your name.

Source Code:

```
#include<stdio.h>
int main(){
    char x[30];
    printf("enter your name:");
    scanf("%[^\n]",x);
    printf("%s",x);
    return 0;
}
```

Output:

```
> cd "h:\Programmi
ng\c\Basic C\exerise class work\1\" ; if ($?) { gcc code18.c -o co
de18 } ; if ($?) { .\code18 }
enter your name: Rashadul islam
 Rashadul islam
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 19

Problem Name: Write a program that read and display a line of text.

Source Code:

```
#include<stdio.h>
int main(){
    char x[30];
    printf("enter your text:");
    scanf("%[^\n]",x);
    printf("your text is:%s",x);
    return 0;
}
```

Output:

```
PS H:\Programming\c\Basic C\exerise class work\1> cd "h:\Programming\c\Basic
 C\exerise class work\1\" ; if ($?) { gcc code19.c -o code19 } ; if ($?) { .
\code19 }
enter your text:this is my lab project and its submission time is coming
your text is:this is my lab project and its submission time is coming
PS H:\Programming\c\Basic C\exerise class work\1>
```

Problem No: 20

Problem Name: Write a program that read any date in the format DD/MM/YYYY and displays day, month and year separately.

Source Code:

```
#include<stdio.h>
int main(){
    int d,m,y;
    printf("Enter the date in the formate DD/MM/YY:");
    scanf("%d/%d/%d",&d,&m,&y);
    printf("day=%d\n",d);
    printf("month=%d\n",m);
    printf("year=%d",y);
    return 0;
}
```

Output:

```
                                        > cd  h:\Programming\c\Basic
 C\exerise class work\1\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter the date in the formate DD/MM/YY:22/12/2004
day=22
month=12
year=2004
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 21

Problem Name: Write a program that read any date in the format DD-MM-YYYY and displays day, month and year separately.

Source Code:

```
#include<stdio.h>
int main(){
    int d,m,y;
    printf("Enter any date in DD-MM-YY:");
    scanf("%d-%d-%d",&d,&m,&y);
    printf("day=%d\n",d);
    printf("month=%d\n",m);
    printf("year=%d",y);
    return 0;
}
```

Output:

```
                                        > cd "h:\Programming\c\Basic
 C\exerise class work\1\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter any date in DD-MM-YY:22-12-2004
day=22
month=12
year=2004
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Problem No: 22

Problem Name: Write a program that read any date in the following format:- i) DD-MM-YYYY ii) DD/MM/YYYY iii) DD MM YYYY iv) DD,MM,YYYY and displays day, month and year separately.

Source Code:

```
#include<stdio.h>
int main(){
    int d,m,y;
    printf("enter the date format DD/MM/YY or
           DD-MM-YY or DD,MM,YY or DD MM YY : ");

    scanf("%d%*[-,/ ] %d%*[-,/ ] %d%*[-,/]",
           &d,&m,&y);
    printf("day=%d\n",d);
    printf("month=%d\n",m);
    printf("year=%d",y);
    return 0;
}
```

Output:

```
PS H:\Programming\c\Basic C> cd "h:\Programming\c\Basic C\exerise class work
\1\" ; if ($?) { gcc code22.c -o code22 } ; if ($?) { .\code22 }
enter the date format DD/MM/YY or DD-MM-YY or DD,MM,YY or DD MM YY : 22 12 2
004
day=22
month=12
year=2004
PS H:\Programming\c\Basic C\exerise class work\1>
```

Daffodil International University

Operator

Problem No:01

Problem Name: Write a program that read two integers and display sum.

Source Code:

```
#include<stdio.h>
int main(){
     int a,b,sum;
     printf("Enter a & b :");
     scanf("%d %d",&a,&b);
     sum= a + b ;
     printf("the sum is:%d",sum);
     return 0;
}
```

Output:

```
                                                > cd "h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a & b :100
200
the sum is:300
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:02

Problem Name: Write a program that subtract two integers.

Source Code:

```c
#include<stdio.h>
int main(){
    int a,b,subtract;
    printf("Enter a & b :");
    scanf("%d %d",&a,&b);
    subtract= a - b ;
    printf("the subtract is:%d",subtract);
    return 0;
}
```

Daffodil International University

Output:

```
> cd  h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a & b :5000
4000
the subtract is:1000
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:03

Problem Name: Write a program that read two integers and display product.

Source Code:

```
#include<stdio.h>
int main(){
    int a,b,product;
    printf("Enter a & b :");
    scanf("%d %d",&a,&b);
    product= a * b ;
    printf("the product is:%d",product);
    return 0;
}
```

Output:

```
                                                    > cd "h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc code3.c -o code3 } ; if ($?) { .\c
ode3 }
Enter a & b :10
50
the product is:500
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:04

Problem Name: Write a program that divide two integers.

Source Code:

```c
#include<stdio.h>
int main(){
    int a,b,div;
    printf("Enter a & b :");
    scanf("%d %d",&a,&b);
    div= a / b ;
    printf("the divided is:%d",div);
    return 0;
}
```

Daffodil International University

Output:

```
> cd "h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a & b :25
5
the divided is:5
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:05

Problem Name: Write a program that read and divide two floating point number.

Source Code:

```
#include<stdio.h>
int main(){
    float a,b,div;
    printf("Enter a & b :");
    scanf("%f %f",&a,&b);
    div= a / b ;
    printf("the divided is:%f",div);
    return 0;
}
```

Output:

```
> cd "h:\Programming\c\Basic C\exerise class work\2\" ; if ($?) { gcc code5.c -o code5 } ; if ($?) { .\code5 }
Enter a & b :509.134
102.876
the divided is:4.949007
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:06

Problem Name: Write a program that read two integer and display remainder.

Source Code:

```
#include<stdio.h>
int main(){
    int a,b,r;
    printf("Enter a & b (a>b):");
    scanf("%d %d",&a,&b);
    r=a%b;
    printf("the reminder is:%d",r);
    return 0;
}
```

Daffodil International University

Output:

```
> cd H:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a & b (a>b):39
7
the reminder is:4
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:07

Problem Name: Write a program that read radius of a circle and display area.

Source Code:

```
#include<stdio.h>
int main(){
    float r,pie,area;
    pie=3.1416;
    printf("Enter radius:");
    scanf("%f",&r);
    area=pie*r*r;
    printf("the area of circle is :%f",area);
    return 0;
}
```

Output:

```
> cd "h:\Programming\c\Basic C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter radius:5
the area of circle is :78.540001
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:08

Problem Name: Write a program that read radius of a circle and display area.

Source Code:

```c
#include<stdio.h>
int main(){
    float r,pie,area;
    pie=3.1416;
    printf("Enter radius:");
    scanf("%f",&r);
    area=pie*r*r;
    printf("the area of circle is :%f",area);
    return 0;
}
```

Daffodil International University

Output:

```
> cd "H:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter radius:6
the area of circle is :113.097595
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:09

Problem Name: Write a program that read radius of a circle and display area.

Source Code:

```c
#include<stdio.h>
int main(){
    float r,pie,area;
    pie=3.1416;
    printf("Enter radius:");
    scanf("%f",&r);
    area=pie*r*r;
    printf("the area of circle is :%f",area);
    return 0;
}
```

Output:

```
                                                           > cd "h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter radius:10
the area of circle is :314.160004
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:10

Problem Name: Write a program that read radius of a circle and display area.

Source Code:

```
#include<stdio.h>
int main(){
    float r,pie,area;
    pie=3.1416;
    printf("Enter radius:");
    scanf("%f",&r);
    area=pie*r*r;
    printf("the area of circle is :%f",area);
    return 0;
}
```

Daffodil International University

Output:

```
                                                  > cd "h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter radius:4.87
the area of circle is :74.509010
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:11

Problem Name: Write a program that read temperature in Celsius and display in Fahrenheit.

Source Code:

```
#include<stdio.h>
int main(){
    float C,F;
    printf("Enter the Celcius temp:");
    scanf("%f",&C);
    F=((9*C/5)+32);
    printf("the temperature in fahrenheit
           is:%f",F);
    return 0;
}
```

Output:

```
> cd "h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter the Celcius temp:39
the temperature in fahrenheit is:102.199997
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:12

Problem Name: Write a program that read temperature in Fahrenheit and display in Celsius.

Source Code:

```
#include<stdio.h>
int main(){
    float C,F;
    printf("Enter the fahrenheit temp:");
    scanf("%f",&F);
    C=((F-32)*5)/9;
    printf("the Calcius temp is:%f",C);
    return 0;
}
```

Output:

```
> cd "h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter the fahrenheit temp:104
the Calcius temp is:40.000000
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No: 13

Problem Name: Write a program that read two numbers and display bitwise AND.

Source Code:

```c
#include<stdio.h>
int main(){
    unsigned int a,b,res;
    printf("Enter a & b :");
    scanf("%u %u",&a,&b);
    res= a & b;
    printf("the bitwise AND num is:%u",res);
    return 0;
}
```

Daffodil International University

Output:

```
> cd "H:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a & b :8
9
the bitwise AND num is:8
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:14

Problem Name: Write a program that read two numbers and display bitwise OR.

Source Code:

```
#include<stdio.h>
int main(){
    int a,b,res;
    printf("Enter a & b :");
    scanf("%d %d",&a,&b);
    res= a | b;
    printf("the bitwise or num is:%d",res);
    return 0;
}
```

Daffodil International University

Output:

```
> cd "h:\Programming\c\Basic
C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a & b :6
4
the bitwise or num is:6
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:15

Problem Name: Write a program that read two numbers and display bitwise Exclusive OR.

Source Code:

```c
#include<stdio.h>
int main(){
    int a,b,res;
    printf("Enter a & b :");
    scanf("%d %d",&a,&b);
    res= a ^ b;
    printf("the bitwise E-OR num is:%d",res);
    return 0;
}
```

Output:

```
> cd  H:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a & b :10
12
the bitwise E-OR num is:6
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:16

Problem Name: Write a program that read a number and divide by two using shift operator.

Source Code:

```
#include<stdio.h>
int main(){
    int a,res;
    printf("Enter a num: ");
    scanf("%d",&a);
    res = a>>1;
    printf("the num divided by 2
is:%d",res);
    return 0;
}
```

Daffodil International University

Output:

```
> cd "h:\Programming\c\Basic C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a num: 9
the num divided by 2 is :4
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:17

Problem Name: Write a program that read a number and multiply by two using shift operator.

Source Code:

```c
#include<stdio.h>
int main(){
    int a,res;
    printf("Enter a num: ");
    scanf("%d",&a);
    res = a << 1;
    printf("the num multiply by 2 is :%d",res);
    return 0;
}
```

Output:

```
> cd "h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a num: 12
the num multiply by 2 is :24
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:18

Problem Name: Write a program that read a number and multiply by five using shift operator.

Source Code:

```
#include<stdio.h>
int main(){
    int a,res;
    printf("Enter a num: ");
    scanf("%d",&a);
    res = (a << 2) + a ;
    printf("the num multiply by 5 is :%d",res);
    return 0;
}
```

Daffodil International University

Output:

```
 C\exerise class work\2\" ; if ($?) { gcc code18.c -o code18 } ; if ($?) { .
\code18 }
Enter a num: 4
the num multiply by 5 is :20
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:19

Problem Name: Write a program that read a number

and mod by 4 using bitwise AND.

Source Code:

```
#include<stdio.h>
int main(){
    int a,res;
    printf("Enter a num:");
    scanf("%d",&a);
    res= a & (4-1);
    printf("number mod by 4 is %d",res);
    return 0;
}
```

Output:

```
                                                        > cd  h:\Programming\c\Basic
 C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRu
nnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a num:5
number mod by 4 is 1
PS H:\Programming\c\Basic C\exerise class work\2>
```

Problem No:20

Problem Name: Write a program that read a number and mod by 8 using bitwise AND.

Source Code:

```c
#include<stdio.h>
int main(){
    int a,res;
    printf("Enter a num:");
    scanf("%d",&a);
    res= a & (8-1);
    printf("number mod by 8 is %d",res);
    return 0;
}
```

Output:

```
> cd "h:\Programming\c\Basic C\exerise class work\2\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a num:44
number mod by 8 is 4
PS H:\Programming\c\Basic C\exerise class work\2>
```

Thank You